

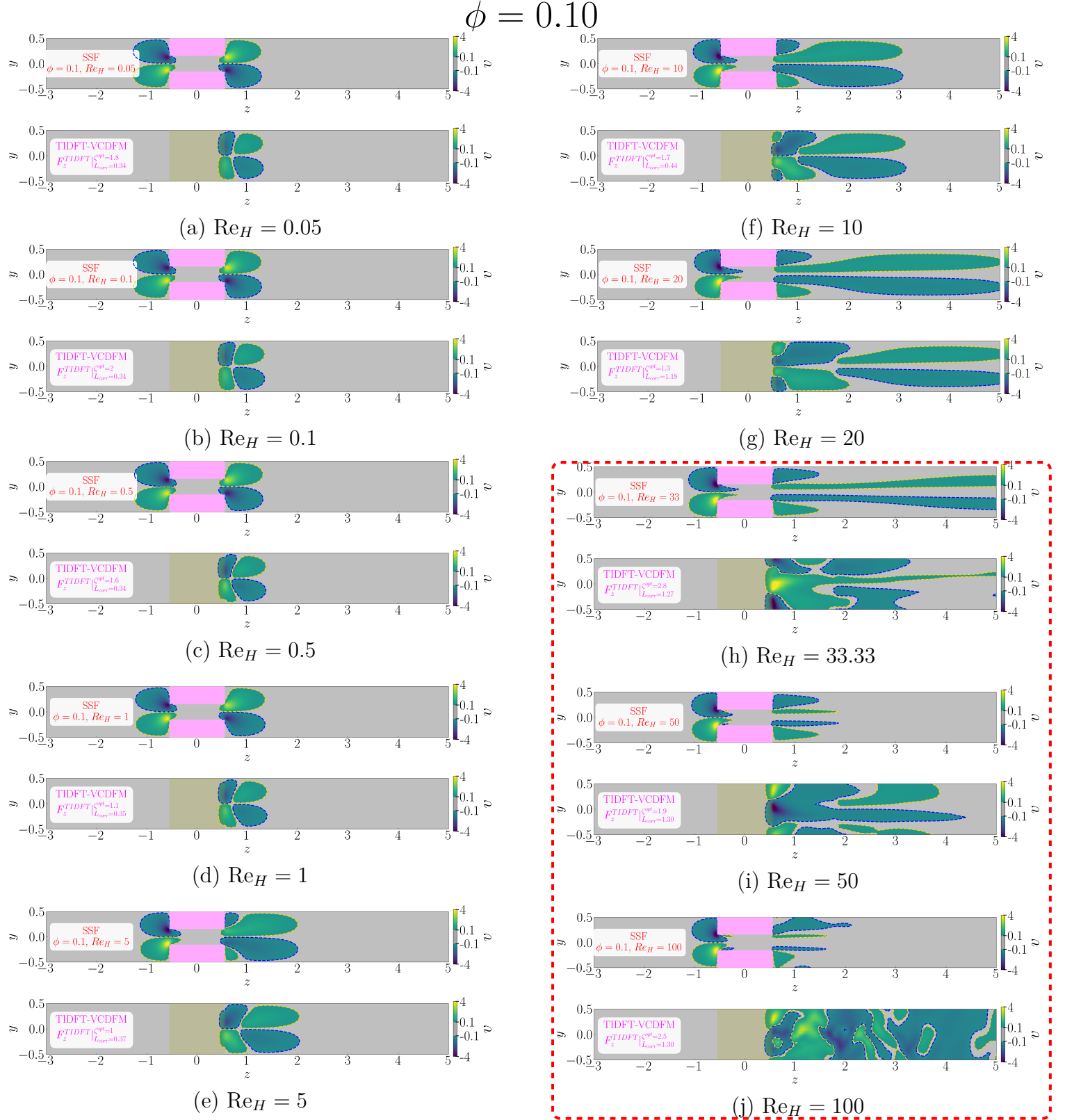


Figure D.4: Transverse velocity component v in the streamwise (z, y) -plane for the SRG configuration at $\phi = 0.10$ and $\alpha = 0^\circ$. For each Reynolds number, the upper panel shows the geometry-resolved small-scale flow (SSF), while the lower panel shows the TIDFT-VCDFM reconstruction using F_z^{TIDFT} . The dashed red rectangle highlights $Re_H = 33.33$, 50, and 100, where noticeable differences between the TIDFT-VCDFM and SSF indicate a loss of predictive accuracy at $\phi = 0.10$.

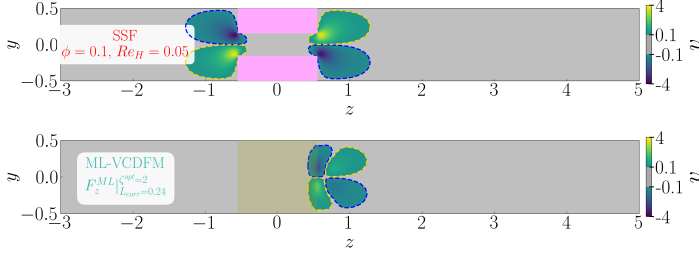
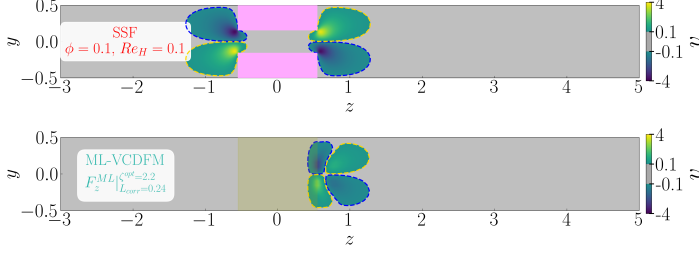
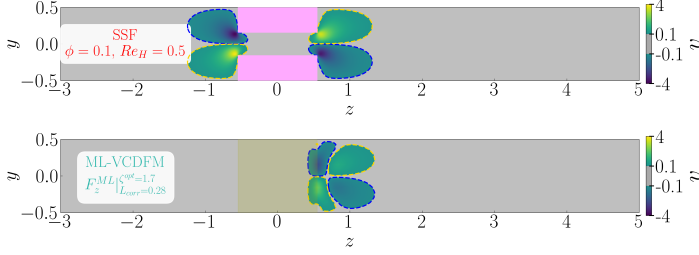
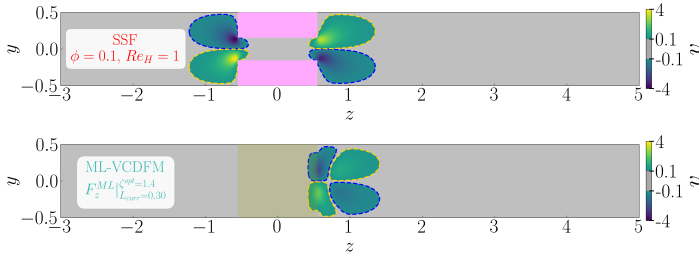
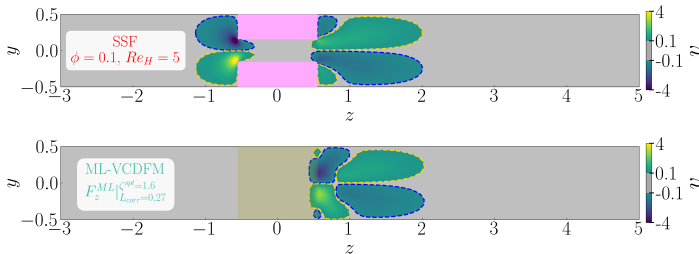
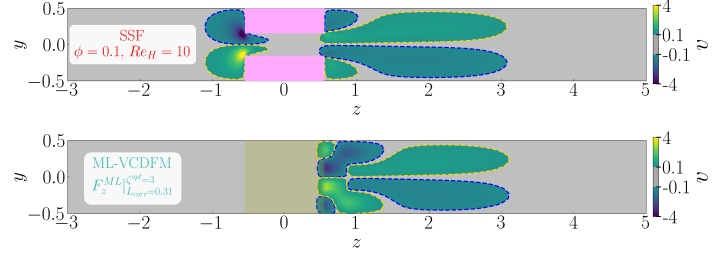
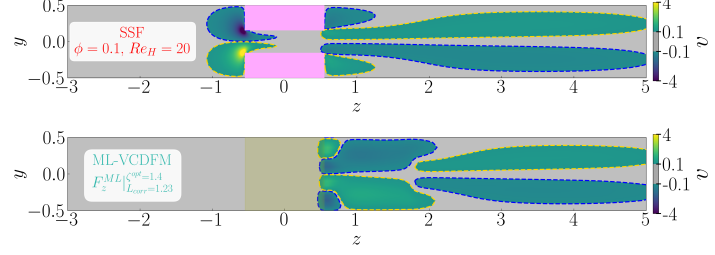
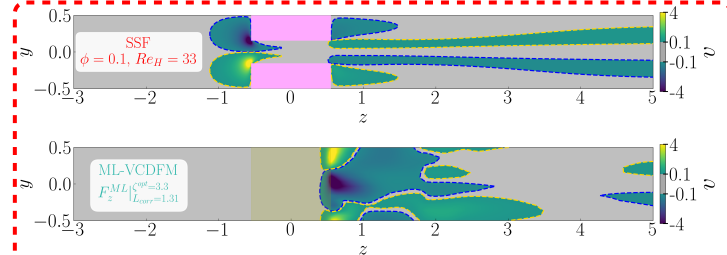
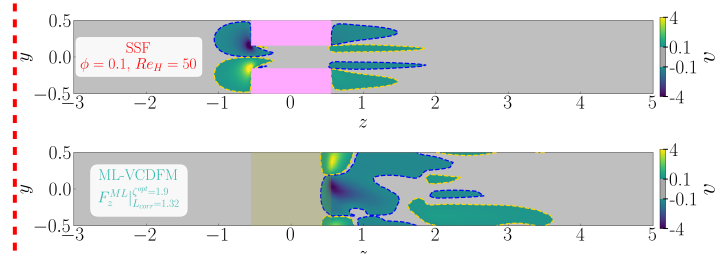
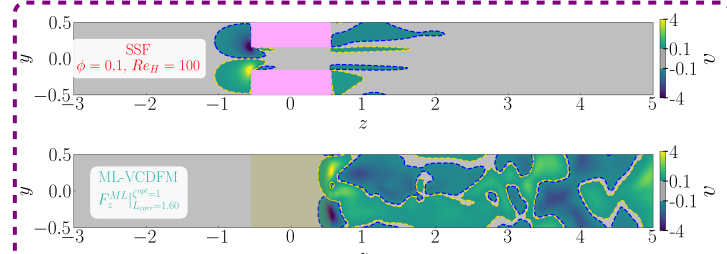
$\phi = 0.10$

(a) $Re_H = 0.05$

(b) $Re_H = 0.1$

(c) $Re_H = 0.5$

(d) $Re_H = 1$

(e) $Re_H = 5$

(f) $Re_H = 10$

(g) $Re_H = 20$

(h) $Re_H = 33.33$

(i) $Re_H = 50$

(j) $Re_H = 100$

Figure D.5: Transverse velocity component v in the streamwise (z,y) -plane for the SRG at $\phi = 0.10$ and $\alpha = 0^\circ$. For each Reynolds number, the upper panel shows the geometry-resolved small-scale flow (SSF), and the lower panel shows the ML-VCDFM reconstruction using F_z^{ML} . The red dashed rectangle marks $Re_H = 33.33$ and 50 , where noticeable differences from the SSF reveal a loss of ML-VCDFM accuracy within the training parameter range. The violet dashed rectangle marks the extrapolation case $Re_H = 100$, where larger differences place the reconstruction outside its effective validity range.

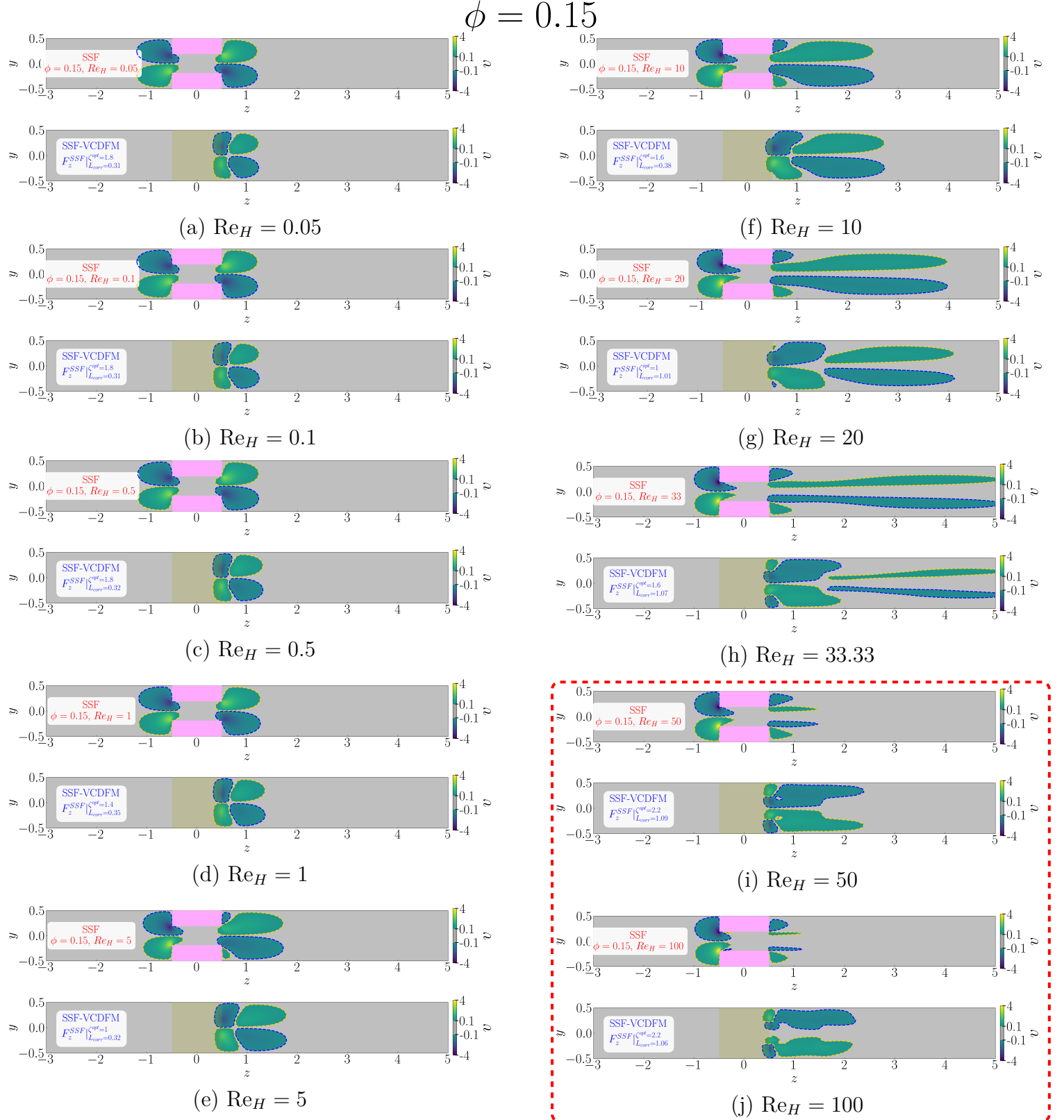


Figure D.6: Transverse velocity component v in the streamwise (z, y) -plane for the SRG at $\phi = 0.15$ and $\alpha = 0^\circ$. For each Reynolds number, the upper panel shows the geometry-resolved small-scale flow (SSF), while the lower panel shows the SSF-VCD FM reconstruction using F_z^{SSF} . The red dashed rectangle marks $Re_H = 50$ and 100 , where noticeable differences between the SSF-VCD FM and SSF indicate a loss of predictive accuracy.

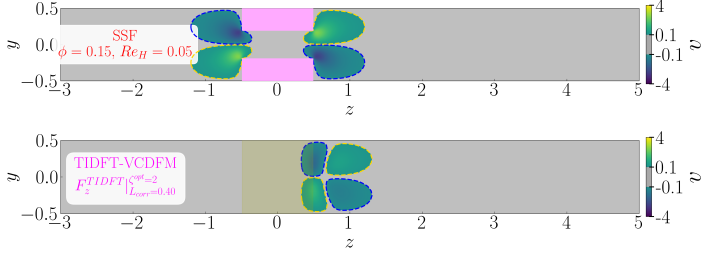
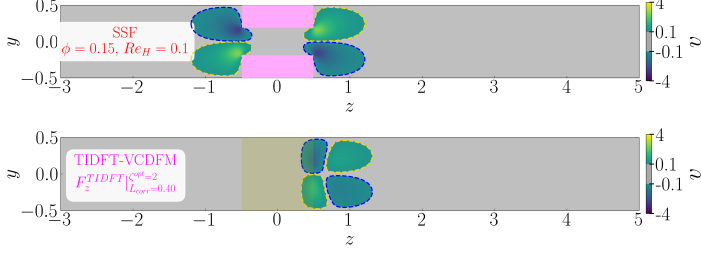
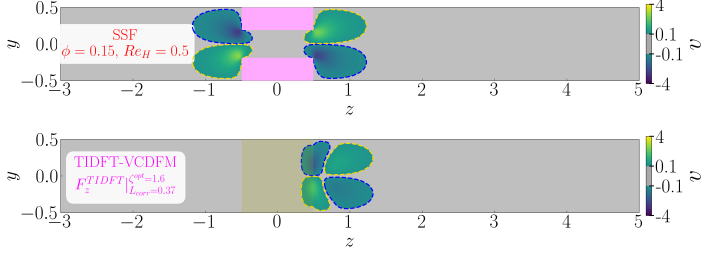
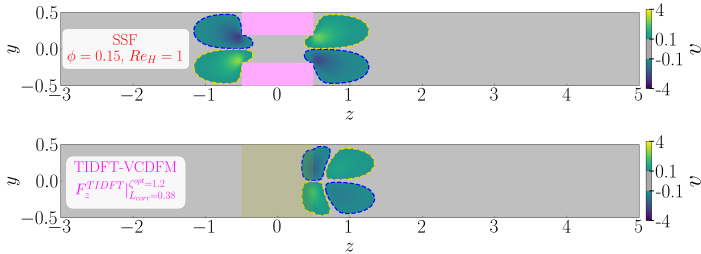
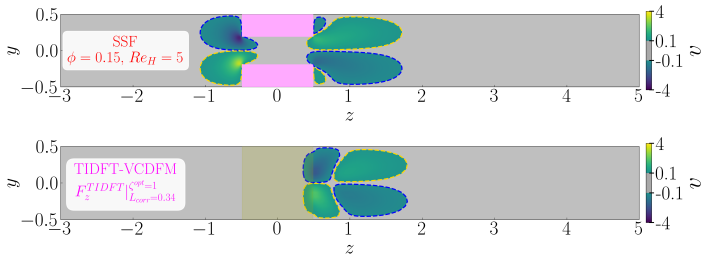
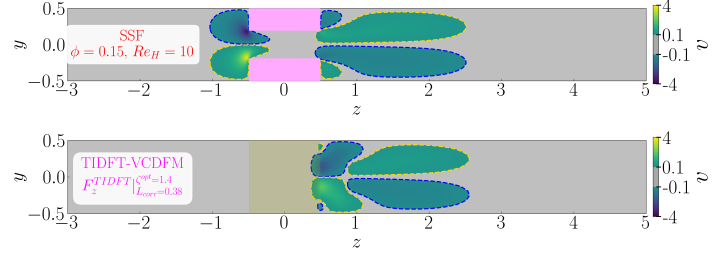
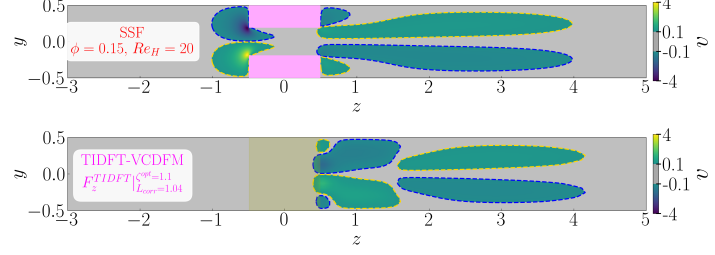
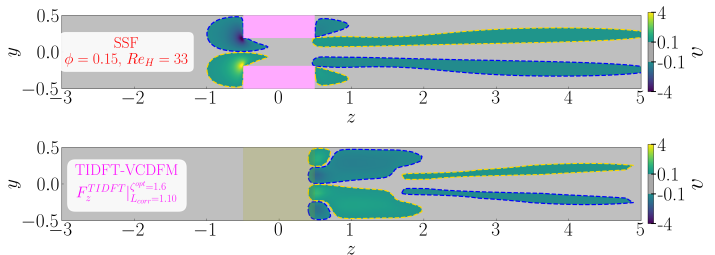
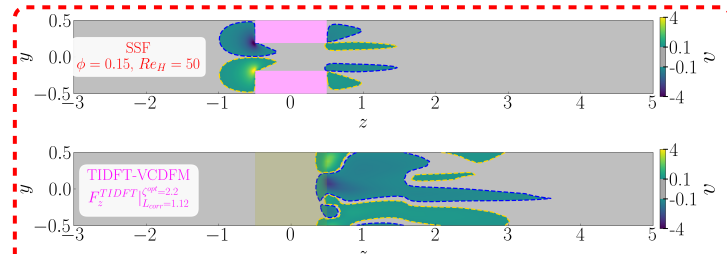
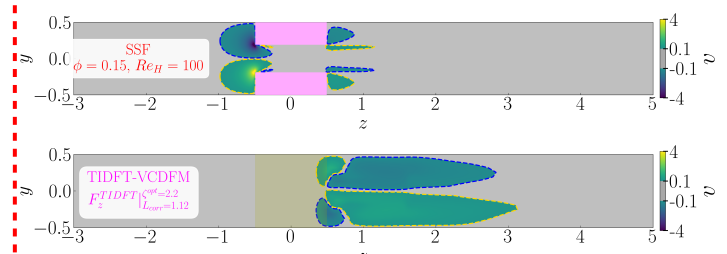
$\phi = 0.15$

 (a) $Re_H = 0.05$

 (b) $Re_H = 0.1$

 (c) $Re_H = 0.5$

 (d) $Re_H = 1$

 (e) $Re_H = 5$

 (f) $Re_H = 10$

 (g) $Re_H = 20$

 (h) $Re_H = 33.33$

 (i) $Re_H = 50$

 (j) $Re_H = 100$

Figure D.7: Transverse velocity component v in the streamwise (z, y) -plane for the SRG at $\phi = 0.15$ and $\alpha = 0^\circ$. For each Reynolds number, the upper panel shows the geometry-resolved small-scale flow (SSF), while the lower panel shows the TIDFT-VCDFM reconstruction using F_z^{TIDFT} . The red dashed rectangle marks $Re_H = 50$ and 100 , where noticeable differences between the TIDFT-VCDFM and SSF indicate a loss of predictive accuracy.